

DSA0112- PROGRAMMING IN C++

Activity-1

1.Arithmetic Operators

QUESTIONS 8 / 48 completed

Arithmetic

ARITHMATIC

Write a program to read in two integers and perform the following operations on them: addition, subtraction, multiplication, division, and modulo.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int a=6,b=8;
5     cout<<"the addition of a and is :"<<(a+b)<<"\n";
6     cout<<"the sub of a and b is :"<<(a-b)<<endl;
7     cout<<"the mul od a and b is :"<<(a*b)<<endl;
8     cout<<"the div od a and b is :"<<(a/b)<<endl;
9     cout<<"the mod of a and b is :"<<(a%b)<<endl;
10    return 0;
11 }
```

2.Factorial

QUESTIONS 8 / 48 completed

Factorial

FACTORIAL

Write a program to read in an integer and print the factorial of that number using a for loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int number;
5     long long fact=1;
6
7     cout<<"enter the number:"<<endl;
8     cin>>number;
9
10    if (number<0){
11        cout<<"the number is less than 0 fact is not possible";
12    }
13    else{
14        for(int i=1;i<=number;i++){
15            fact=fact*i;
16        }
17        cout<<"the factorail of the number is :"<<fact;
18    }
19    return 0;
20 }
21 }
```

INPUT

3

3.PrimeNumbers

QUESTIONS 8 / 48 completed

Prime

PRIME

Write a program to read in an integer and determine if it is a prime number or not using a for loop.

PROGRAM EDITOR

C++

Run

Save

INPUT

7

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4
5     int number;
6     bool isPrime=true;
7
8     cout<<"enter number:"<<endl;
9     cin>>number;
10
11     if(number<=1){
12         isPrime=false;
13     }
14     else{
15         for(int i=2;i<number;i++){
16             if(number%i==0){
17                 isPrime = false;
18                 break;
19             }
20         }
21     }
```

4.Fibonacci

QUESTIONS 8 / 48 completed

Fibonacci

FIBONACCI

Write a program to read in an integer and print the Fibonacci series up to that number using a for loop.

PROGRAM EDITOR

C++

Run

Save

INPUT

6

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int number;
5     int a=0,b=1,c;
6
7     cout<<"enter the number of terms:"<<endl;
8     cin>>number;
9
10    cout<<"the fibonacci series are:";
11
12    for (int i=1;i<=number;i++){
13        cout<<a<<" ";
14        c=a+b;
15        a=b;
16        b=c;
17    }
18    return 0;
19 }
```

5.GCD

QUESTIONS 8 / 48 completed

GCD

GCD

Write a program to read in two integers and compute their greatest common divisor using a do-while loop.

PROGRAM EDITOR

C++

Run

Save

INPUT

2 5

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int a,b,remainder;
5
6     cout<<"enter two number";
7     cin>>a>>b;
8
9     if (b=0){
10         cout<<"gcd is:"<<a;
11         return 0;
12     }
13     do{
14         remainder=a%b;
15         a=b;
16         b=remainder;
17     }while(b!=0);
18     cout<<"the gcd is : "<<a;
19     return 0;
20 }
```

6.Palindrome

QUESTIONS 8 / 48 completed

Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++

Run

Save

INPUT

121

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int number,original,digit,reverse=0;
5
6     cout<<"enter the number:"<<endl;
7     cin>>number;
8
9     original=number;
10
11     for(;number>0; number=number/10){
12         digit=number%10;
13         reverse = reverse*10+digit;
14     }
15     if (original==reverse)
16         cout<<"it is palindrome"<<original;
17     else
18         cout<<"it is not palindrome";
19     return 0;
20 }
```